Nikhil Kolhe

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Education

Dr. D.Y. Patil Institute Of Technology, Pune. June 2024
B.E.(A.I. And Data Science); CGPA:8.5 *Pune, Maharashtra*

Skills

Programming Languages: Python, SQL

Generative AI & Agentic AI: LLMs, LangChain, LangGraph, MCP, FastMCP, Multi-Agent Systems, RAG, Embeddings, Vector Search, Prompt Engineering, Tool Calling, Structured Outputs, Human-in-the-Loop Workflows

LLM Providers & APIs: OpenAI, Google Gemini, Groq, REST APIs, FastAPI, Flask, GraphQL

Databases & Deployment: SQLite, MySQL, PostgreSQL, Snowflake, SQLAlchemy, Chroma, FAISS, Git/GitHub

Machine Learning, Data & BI: scikit-learn, Pandas, NumPy, Deep Learning (RNN/LSTM), NLP, OCR, Databricks, Power BI, Tableau, TIBCO Spotfire

Experience

I2E Consulting 2024 – Present
Associate Developer *Remote*

AI-Powered Resume Analyzer | *Python, Streamlit, Gemini, Embeddings, PyPDF2, Tesseract OCR*

- Built an LLM-powered application to extract resume content from multiple file formats, including scanned documents via OCR, generate structured feedback, and compare resumes against job descriptions using embedding similarity.

Clinical Study Assistant Agent | *Python, LangGraph, FastMCP, FastAPI, scikit-learn*

- Integrated trained KPI prediction models as MCP tools to predict clinical study KPIs, retrieve similar studies, and compare study results. Developed a LangGraph agent that interprets natural-language queries, selects appropriate tools, and returns results through REST APIs.

Time Card Report | *Power BI, Power Automate, Data Gateway*

- Developed interactive Power BI reports to analyze employee working hours, overtime, project allocation, resource utilization, and time-reporting compliance. Created an executive infographic report presenting key workforce insights for management and supporting data-driven decisions.

Projects

Drug Discovery Agent | *Python, LangGraph, MCP, FastAPI, OpenAI, Gemini, RAG*

- Built a multi-agent drug-target prioritization system that integrates biomedical evidence from DepMap, Open Targets, Pharos, and Europe PMC using MCP-connected data tools. Implemented a stateful LangGraph workflow for gene validation, parallel evidence collection, data normalization, evidence scoring, and cross-source conflict detection. Developed evidence-grounded research report generation and RAG-based follow-up queries, with source-traceable results, human-in-the-loop review, and FastAPI endpoints.

AI-Powered Blog Generator (LLM) | *Llama 3.1 8B, AWS Bedrock, RAG, Streamlit*

- Developed an AI-powered content generator leveraging Llama 3.1 8B, AWS Bedrock, and RAG for real-time fine-tuning on dynamic datasets. Integrated with Streamlit for a seamless user interface, the system automated blog creation, editing, and publishing, increasing efficiency in content marketing and engagement for enterprises.

Clinical Study Resource Demand Forecasting (Pfizer) | *Databricks, PySpark, Delta Lake, MLflow, XGBoost/LightGBM, Databricks Work*

- Built and deployed a production regression model on Databricks to predict study-wise monthly resource demand for Pfizer clinical studies, enabling proactive staffing and capacity planning across the study portfolio. Engineered features in PySpark on Delta Lake from study attributes, timelines, and historical resource allocation, tracked experiments in MLflow, and automated monthly inference and retraining with Databricks Workflows, writing forecasts to Delta tables for BI dashboards. Added data-drift and forecast-accuracy monitoring, improving forecast accuracy by [X]% over the baseline.

Certifications & Publications

AWS Certified AI Practitioner (AIF-C01)
IEEE Publication: *Fun Soundify: Music Generation powered by AI*